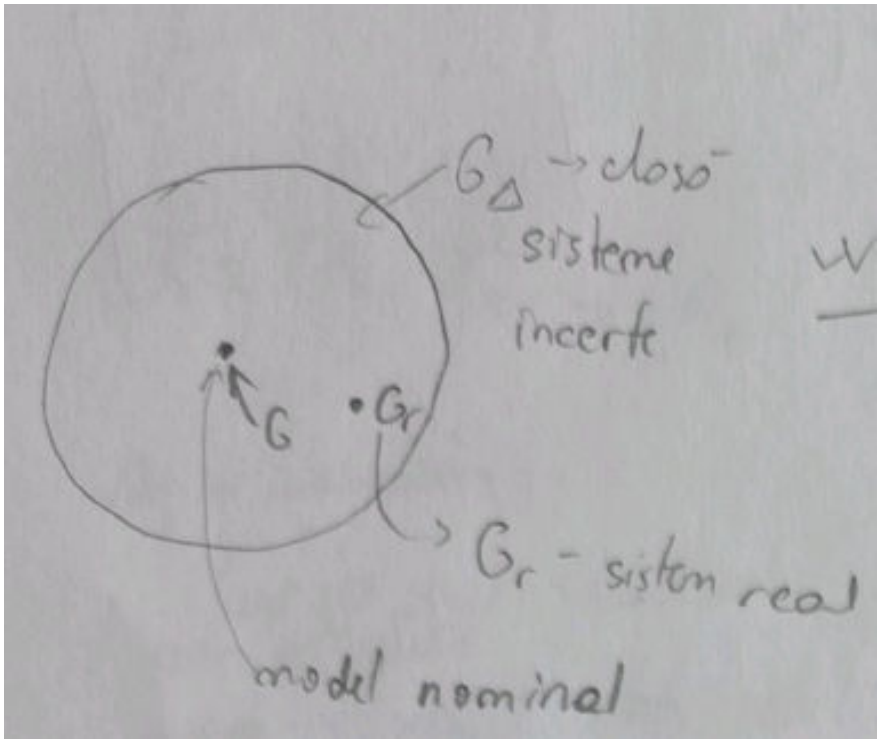
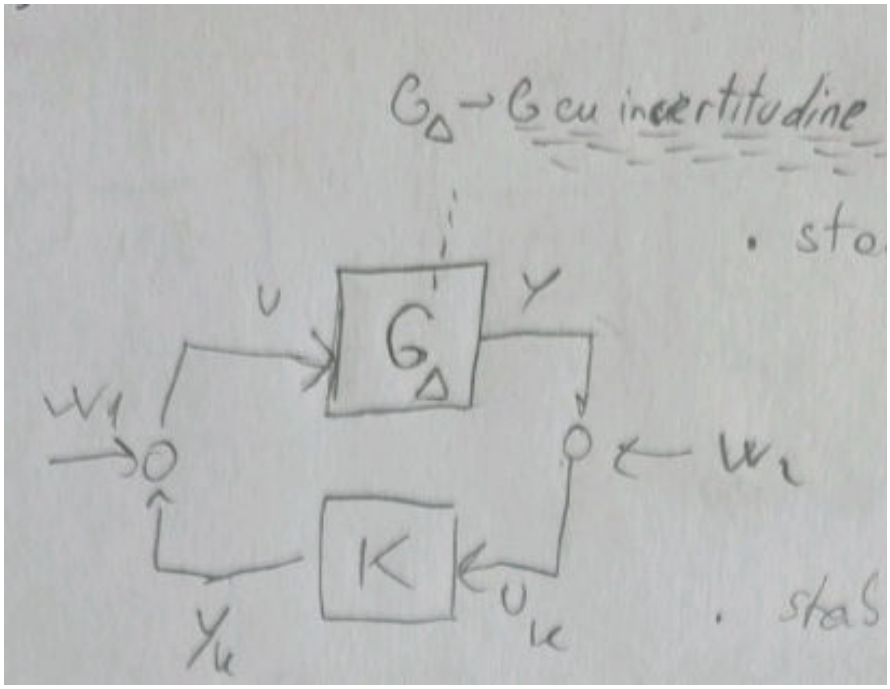
RRA

LAB 4 (Parteneriat cu laburile scrise de PETrascu)

Fie $G : \begin{cases} \dot{x} = \mathrm{A}x + \mathrm{B}u \\ y = \mathrm{C}x + \mathrm{D}u \end{cases}$. Presupunem G_Δ ca fiind G cu incertitudini



Fie sistemul in bucla de reactie



Stabilitatea pe stare $\to \Lambda(A_{\text{cl}}) \in \mathbb{C}_-$ este echivalenta cu stabilitatea I/O $\to T_{\text{cl}}(s) \in \mathscr{RH}_\infty \Leftrightarrow K, G$ stabilizabile si detectabile

Ce vrem: sa construim un regulator / compensator K care stabilizeaza tot G_Δ.

O matrice U se numeste **unitara** daca $\begin{cases} \text{UU}^* = I \\ ||A||_{\{2,\infty\}} = ||\text{UA}||_{\{2,\infty\}} = ||\text{AU}||_{\{2,\infty\}} \end{cases}$

Acum vine o intrebarte fireasca: aceste incertitudini cum pot fi modelate? Exista 3 moduri:

1. **In mod aditiv:** $G_\Delta(s) = G(s) + \Delta_a(s)$, unde $\Delta_a(s)$ sunt incertitudinile sau $G_\Delta(s) = G(s) + \gamma\widetilde{\Delta}_a(s)$, cu $||\widetilde{\Delta}_a(s)||_\infty < 1$ (se preferea asta). **Putem lua aceasta norma infinit si 1 pentru a ne usura calculele**. Presupuneri despre aceste incertitudini: $\begin{cases} \Delta_a(s) \in \mathscr{RH}_\infty \\ ||\Delta_a(s)||_\infty < \gamma, \gamma > 0 \end{cases}$
2. **In mod multiplicativ:** $G_\Delta(s) = G(s)(1 + W_T\Delta)$, unde $||\Delta||_\infty \leq 1$. Asa se nota la **TSA.** In realitate avem: I**ncertitudini pe** $\begin{cases} \text{iesire} : G_{\Delta_o}(s) = (I_p + \Delta_{\text{no}}(s))G(s) \\ \text{intrare} : G_{\Delta_i}(s) = G(s)(I_m + \Delta_{\text{ni}}(s)) \end{cases}$. Avem alte presupuneri: $G \in \mathbb{R}^{\text{pxm}}$, $\begin{cases} \Delta_{\text{no}}, \Delta_{\text{ni}} \in (\mathscr{R})\mathscr{H}_\infty \\ ||\Delta_{\text{no}}(s)||_\infty < \gamma_{\text{no}}, ||\Delta_{\text{ni}}(s)||_\infty < \gamma_{\text{ni}} \end{cases}$. *Am pus paranteza rotunda la $\mathscr{R}$ deoarece incertitudinile nu*

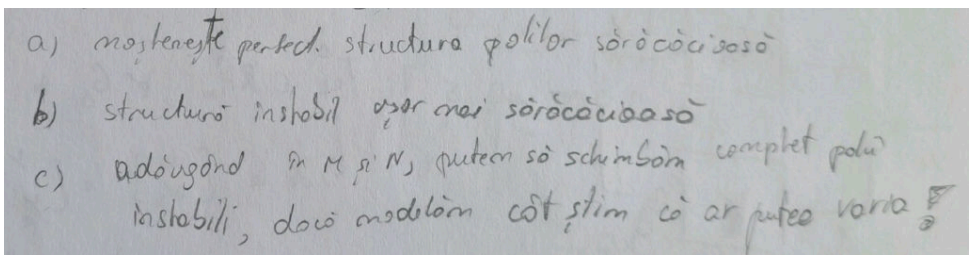
trebuie sa fie neaparat rationale, ci doar stabile. De exemplu, exponentiala: $\Delta(s) = e^{-\tau s}$. $\sup_w |e^{-\mathrm{jws}}| = \sup_w |\cos(\mathrm{ws}) - \mathrm{jsin}(\mathrm{ws})| = 1$. Aceasta reprezentare a incertitudinilor, cea multiplicativa, se prefera la TSA deoarece conditiile de robustete se scriu mai ok

3. **Pe factori coprimi (NU NEAPARAT DUBLU COPRIMI):** $G = \mathrm{NM}^{-1} = \widetilde{M}^{-1}\widetilde{N}$ factorizare la dreapta respectiv stanga, cu $N, M, \widetilde{N}, \widetilde{M} \in \mathscr{RH}_\infty$. $\exists X, Y, \widetilde{X}, \widetilde{Y} \in \mathscr{RH}_\infty$

a.i $\begin{cases} \begin{bmatrix} \widetilde{Y} & -\widetilde{X} \end{bmatrix} \begin{bmatrix} M \\ N \end{bmatrix} = \widetilde{Y}M - \widetilde{X}N = I \left(\text{Bezout la dreapta} \right) \\ \begin{bmatrix} \widetilde{N} & \widetilde{M} \end{bmatrix} \begin{bmatrix} -X \\ Y \end{bmatrix} = \widetilde{M}Y - \widetilde{N}X = I \left(\text{Bezout la stanga} \right) \end{cases}$. Incertitudinile pot aparea pe:

$$\begin{cases} \text{stanga} : G_{\Delta\text{ecf}} = \left(\widetilde{M} + \Delta_{\widetilde{M}} \right)^{-1} \left(\widetilde{N} + \Delta_{\widetilde{N}} \right) \\ \text{dreapta} : G_{\Delta\text{rcf}} = (N + \Delta_N)^{-1}(M + \Delta_M) \end{cases}, \text{ uinde } \begin{cases} \begin{bmatrix} \Delta_{\widetilde{M}} & \Delta_{\widetilde{N}} \end{bmatrix} \in \mathscr{RH}_\infty \text{ cu } \left\| \begin{bmatrix} \Delta_{\widetilde{M}} & +- \Delta_{\widetilde{N}} \end{bmatrix} \right\|_\infty < \gamma_e \\ \begin{bmatrix} \Delta_M \\ \Delta_N \end{bmatrix} \in \mathscr{RH}_\infty \text{ cu } \left\| \begin{bmatrix} \Delta_M \\ \Delta_N \end{bmatrix} \right\|_\infty < \gamma_r \end{cases}$$

Din primele 2 tipuri de modelari de incertitudini, se poate observa ca G_Δ mosteneste polii instabili ai lui G.


a) moștenește perfect structura polilor sărăcăcioasă
b) structură instabilă ușor mai sărăcăcioasă
c) adăugând în M și N, putem să schimbăm complet polii instabili, dacă modelăm cât știm că ar putea varia

Sa luam un exemplu numeric pentru a intelege mai exact

Fie sistemul **nominal** $G(s) = \dfrac{1}{s+1}$ si cel **real** $\widetilde{G}(s) = \dfrac{s-2}{(s+1)(s-3)}$. Putem gasi G_Δ a.i $\widetilde{G} \in G_\Delta$

Mai intai sa folosim **incertitudini aditive:**

$G_\Delta(s) = G(s) + \Delta_a(s) \Rightarrow \Delta_a(s) = G_\Delta(s) - G(s) = \dfrac{(s-2)-(s-3)}{(s+1)(s-3)} = \dfrac{1}{(s+1)(s-3)}$ care **NU este stabil** $(\notin \mathscr{RH}_\infty)$

```
clear
close all
clc

s = tf('s');

G = 1 / (s + 1);
Gtilde = (s - 2) / ((s + 1) * (s - 3));

Delta_a = Gtilde - G
```

```
Delta_a =

         s + 1
  -------------------
  s^3 - s^2 - 5 s - 3

Continuous-time transfer function.
Model Properties
```

```
disp('Este sistemul stabil?')
```

```
Este sistemul stabil?
```

```
disp(roots(Delta_a.den{1}))
```

```
   3.0000 + 0.0000i
  -1.0000 + 0.0000i
  -1.0000 - 0.0000i
```

Trebuie sa fie 3 si -1. Faza aici este ca nu au simplificat pe $s+1$ la numarator si numitor

Acum folosim **incertitudini multiplicative:**

$\widetilde{G}(s) = G(s)(1 + W_T\Delta)$, unde $||\Delta||_\infty \leq 1$. Aici luam norma infinit 1 pentru calcule mai simple. In acest sens, avem $W_T = \dfrac{1}{s-3} \notin \mathscr{RH}_\infty$

```
clear
close all
clc

s = tf('s');

G = 1 / (s + 1);
Gtilde = (s - 2) / ((s + 1) * (s - 3));

Wt = Gtilde / G - 1
```

```
Wt =

      s + 1
  -------------
  s^2 - 2 s - 3

Continuous-time transfer function.
Model Properties
```

```
disp('Este sistemul stabil?');
```

```
Este sistemul stabil?
```

```
disp(roots(Wt.den{1}))
```

```
    3.0000
   -1.0000
```

Aici la fel, nu este facuta simplificarea

Acum folosim **incertitudini pe factori coprimi:**

$G = \frac{N}{M}, N \text{ si } M \text{ corpime}$. Pentru sistemul real, avem $\widetilde{G} = \frac{\widehat{N}}{\widehat{M}}$, $\widehat{N} \text{ si } \widehat{M}$ nu neaparat coprime

In acest sens, alegem $\begin{cases} N = G \\ M = 1 \end{cases}$. Asadar, avem: $\widetilde{G} = \frac{N + \Delta_N}{M + \Delta_M} = \frac{\widehat{N}}{\widehat{M}} \Rightarrow$ $\Delta_N = \hat{N} - N$ $\Delta_M = \hat{M} - M$. Acum alegem $\widehat{N}, \widehat{M} \begin{cases} \widehat{N} = \frac{s-2}{(s+1)^2} \\ \widehat{M} = \frac{s-3}{s+1} \end{cases}$. Zerouri: $\begin{cases} \widehat{N} : \text{in } 2 \text{ si la } \infty \\ \widehat{M} : \text{in } 3 \end{cases}$. Zerouri instabile diferite, deci factorizare coprima. Acum scoatem

$$\Delta_N = \hat{N} - N = \frac{s-2}{(s+1)^2} - \frac{1}{s+1} = \frac{-3}{(s+1)^2}$$

$$\Delta_M = \hat{M} - M = \frac{s-3}{s+1} - 1 = \frac{-4}{s+1}$$

$\Delta_{\text{cf}} = \begin{bmatrix} \Delta_N \\ \Delta_M \end{bmatrix} = \begin{bmatrix} \frac{-3}{(s+1)^2} \\ \frac{-4}{s+1} \end{bmatrix} \in \mathscr{RH}_\infty$. Daca calculam norma infinit, obtinem

$||\Delta_{\text{cf}}||_\infty = ||\begin{bmatrix} \frac{-3}{(s+1)^2} \\ \frac{-4}{s+1} \end{bmatrix}||_\infty = \sqrt{(-3)^2 + (-4)^2} = 5$ (Pitagora)

N si M se pot lua cum vrem.

Remarca: un **avantaj** al **perturbatiilor pe factori coprimi** este ca rezulta intotdeauna **stabile** (ca diferente de 2 MT din $\mathscr{RH}_\infty$)

```
clear
close all
clc

s = tf('s');

G = 1 / (s + 1);

N = G;
M = 1;


N_hat = (s - 2) / (s + 1)^2;
M_hat = (s - 3) / (s + 1);

Delta_N = N_hat - N;
Delta_M = M_hat - M;

disp('Delta_N:');
```

```
Delta_N:
```

```
Delta_N
```

```
Delta_N =

        -3 s - 3
  ---------------------
  s^3 + 3 s^2 + 3 s + 1

Continuous-time transfer function.
Model Properties
```

```
disp('Delta_M:');
```

```
Delta_M:
```

```
Delta_M
```

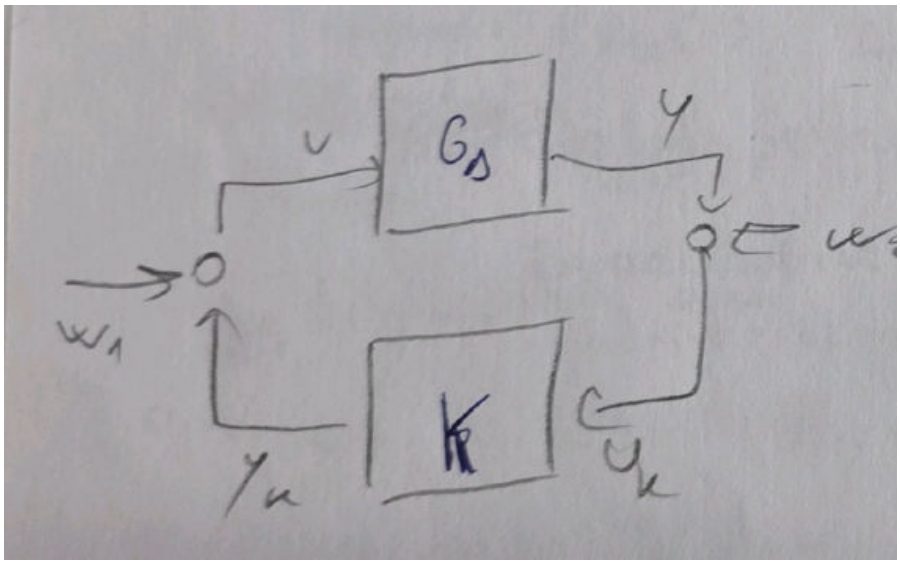
```
Delta_M =

   -4
  -----
  s + 1

Continuous-time transfer function.
Model Properties
```

$$\mathbf{T}^a = \left[\begin{array}{c|c} \mathbf{T}^a_{11} & \mathbf{T}^a_{12} \\ \hline \mathbf{T}^a_{21} & \mathbf{T}^a_{22} \end{array}\right] = \left[\begin{array}{c|c} O & I_m \\ \hline I_p & \mathbf{G} \end{array}\right],$$

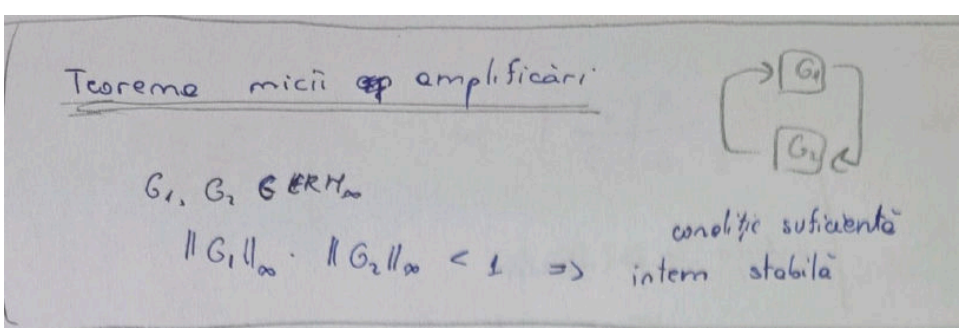
$$\mathbf{T}^m = \left[\begin{array}{c|c} \mathbf{T}^m_{11} & \mathbf{T}^m_{12} \\ \hline \mathbf{T}^m_{21} & \mathbf{T}^m_{22} \end{array}\right] = \left[\begin{array}{c|c} O & \mathbf{G} \\ \hline I_p & \mathbf{G} \end{array}\right],$$

$$\mathbf{T}^{cf} = \left[\begin{array}{c|c} \mathbf{T}^{cf}_{11} & \mathbf{T}^{cf}_{12} \\ \hline \mathbf{T}^{cf}_{21} & \mathbf{T}^{cf}_{22} \end{array}\right] = \left[\begin{array}{c|c} O & I_m \\ \widetilde{\mathbf{M}}^{-1} & \mathbf{G} \\ \hline \widetilde{\mathbf{M}}^{-1} & \mathbf{G} \end{array}\right].$$

Fie sistemul in bucla de reactie

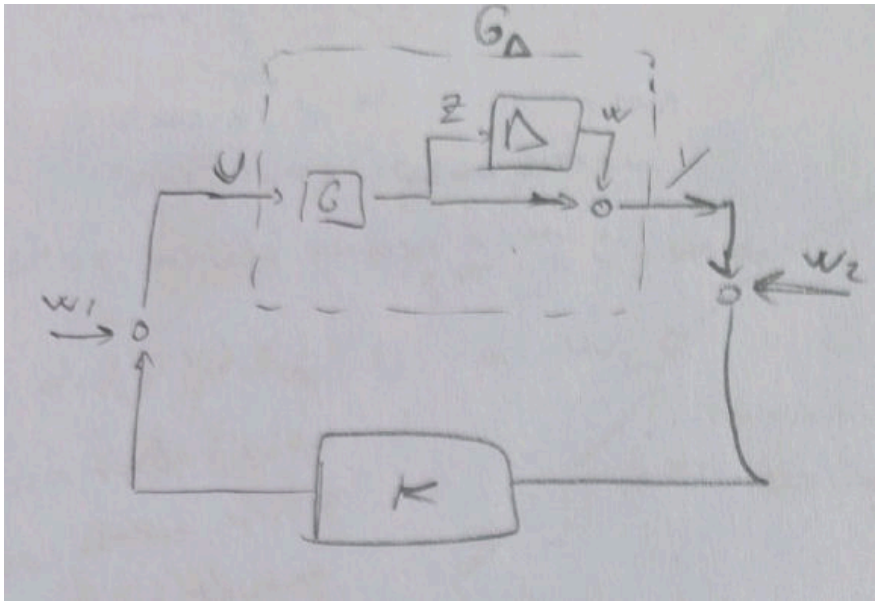


$T^{\Delta}_{\text{cl}} \in \mathscr{RH}_\infty, \forall G_\Delta$ in clasa mea, unde $\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = T^{\Delta}_{\text{cl}} \begin{bmatrix} y \\ y_K \end{bmatrix}$. Presupunem ca G_Δ este stabilizabil si detectabil. Bine, oricum e echivalent cu stabilizare I/O.

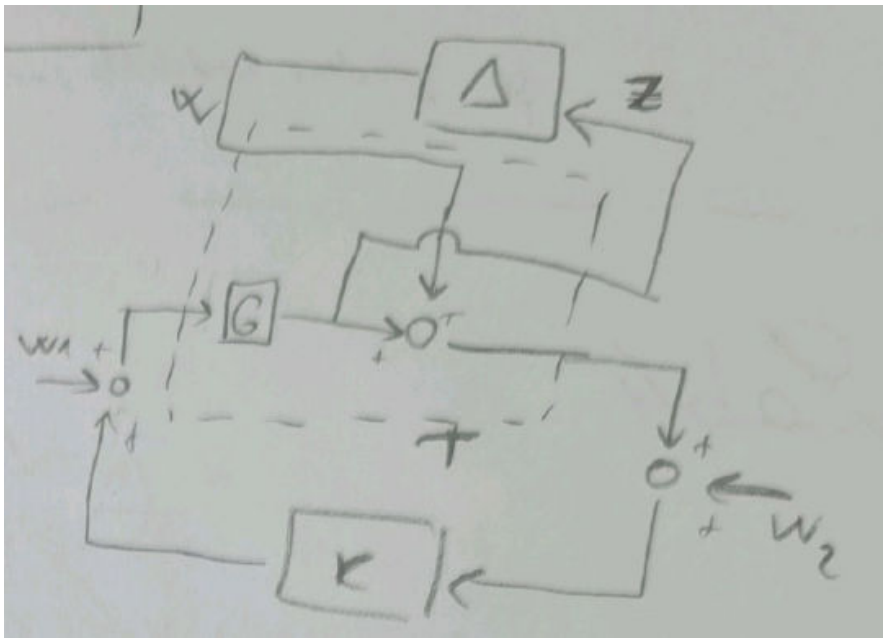


https://en.wikipedia.org/wiki/Small-gain_theorem

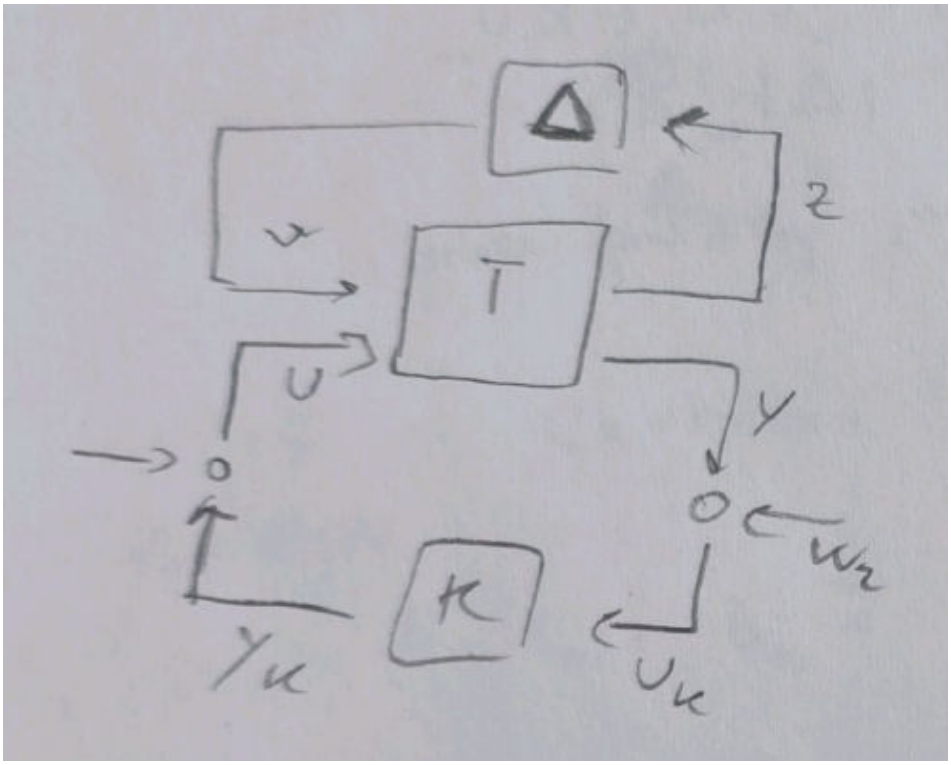
Stim ca $G_\Delta = (I + \Delta)G$. Cum arata asta pe sceme bloc?



De aici rescriem



Forma finala:



$\begin{bmatrix} z \\ y \\ y_K \end{bmatrix} = \widehat{T}_{\text{cl}} \begin{bmatrix} w \\ w_1 \\ w_2 \end{bmatrix}$, unde $\widehat{T}_{\text{cl}}$ stabil in configuratie de TLFi ($\in \mathscr{RH}_\infty$). Noi dorim sa gasim un K astfel incat

$$\begin{cases} \text{sa stabilizeze in TLFi pe } T \\ ||\text{TLFi}(T, K)||_\infty < \dfrac{1}{||\Delta||_\infty} \end{cases}$$

Asta ne spune ca toate cele 3 tipuri de incertitudini se scriu ca un $\text{TLFs}(T, \Delta)$**, unde**

$$\begin{cases} T \text{ este sistemul in bucla inchisa (sistem generalizat)} \\ \Delta \text{ este eroarea (perturbatia) corespunzatoare} \end{cases}$$

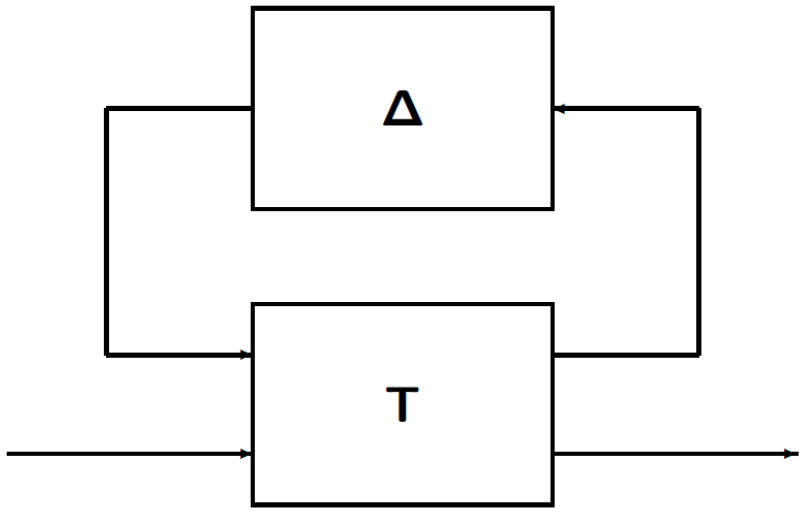
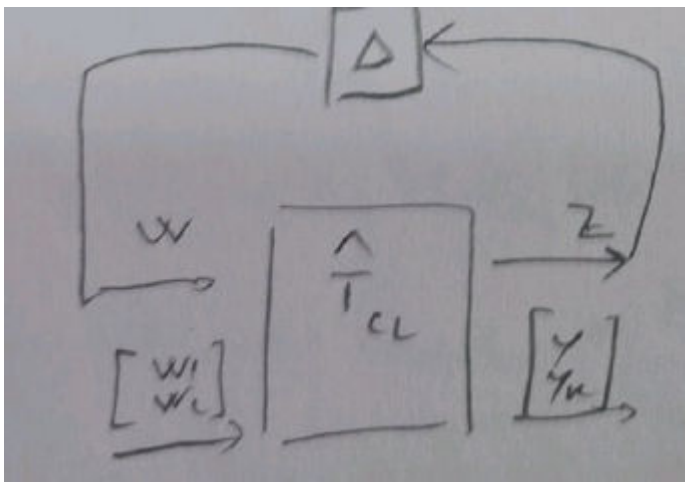
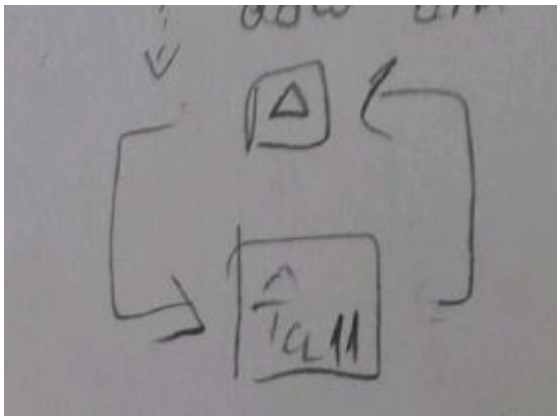


Fig. 12: Modelul sistemelor incerte $\mathbf{G_\Delta} = \mathcal{F}_u(\mathbf{T}, \mathbf{\Delta})$.



Din cate se poate oberva, este ca si cum am conecta pe Δ la **prima intrare si prima iesire**

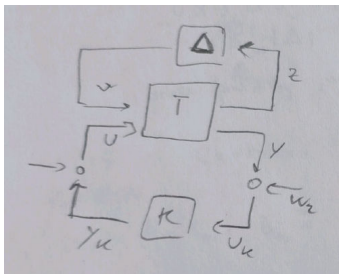


Stim ca $||\widehat{T}_{\mathrm{cl}}||_\infty * ||\Delta||_\infty < 1$, deci Δ stabilizeaza in b.i pe T_{cl}

In acest sens, vrem

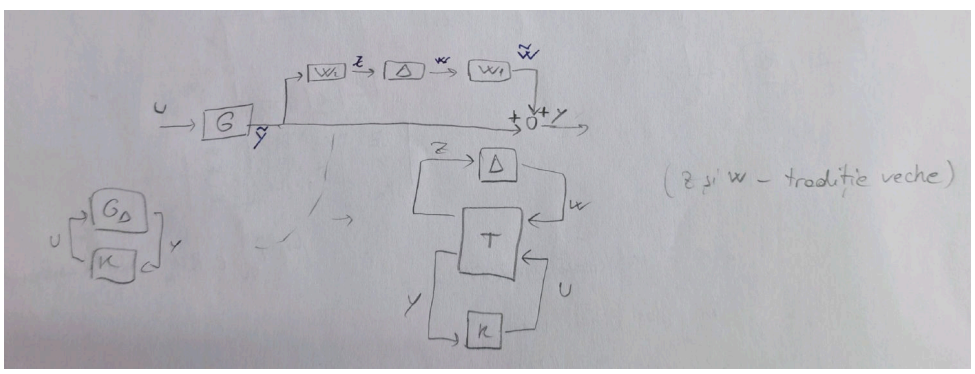
- $G_\Delta = \text{TLFs}(T, \Delta)$
- $T = \text{TLFi}(G, K)$

Esenta este sa gasim $K = ?$ a.i $\text{TLFs}(T, \Delta)$ stabil. O metoda ar fi sa folosim **Teorema Amplificarilor Mici,** dar este o conditie nenecesara si suficienta. Ideea faptului ca noi luam cel mai rau caz $\forall \Delta$ ne asigura teoretic ca

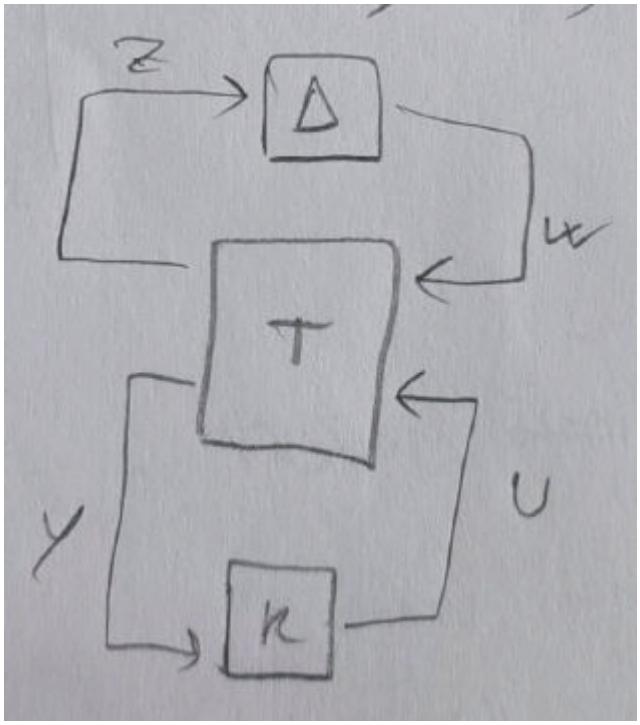


este stabil

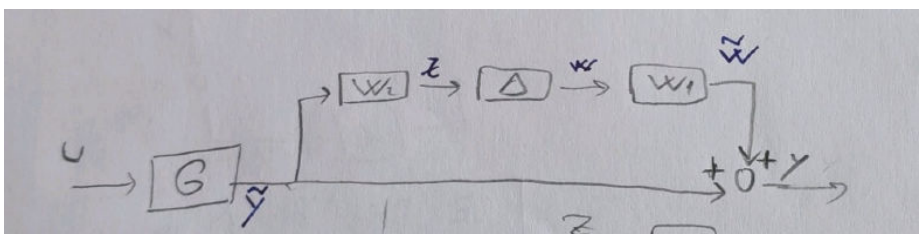
Fie sistemul pe caz general pe incertitudine: $G_\Delta = (I + W_1 \Delta W_2) G = \text{TLFs}(G, \Delta)$, unde $W_1, W_2, \Delta \in \mathscr{RH}_\infty$, iar $||\Delta||_\infty < 1$. In general, incertitudinile apar la frecvente mari, deci W_1, W_2 sunt benzi trece-sus



Pentru a afla T ala, avem 2 variante: *una mai grea*, si anume "trasul de sarme", dar *mai simplu* este sa scoatem forme minimale pentru G, W_1, W_2. Fie $\begin{cases} G = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \\ W_1 = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \\ W_2 = \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} \end{cases}$, realizari minimale.



Uitandu-ne aici, avem: $\begin{bmatrix} z \\ y \end{bmatrix} = T \begin{bmatrix} w \\ u \end{bmatrix}$. Daca scriem ecuatiile:



$$\begin{cases} y = \mathrm{Gu} + W_1 w = \widetilde{y} + W_1 w \\ z = W_2 \widetilde{y} = W_2 \mathrm{Gu} \\ \widetilde{y} = \mathrm{Gu} \end{cases}$$

Ne uitam la G

$$\begin{cases} \dot{x} = \mathrm{Ax} + \mathrm{Bu} \\ \widetilde{y} = \mathrm{Cx} + \mathrm{Du} \end{cases}$$

Acum ne uitam la W_2

$$\begin{cases} \dot{x_{w_2}} = A_{w_2} x_{w_2} + B_{w_2} \widetilde{y} \\ z = C_{w_2} x_{w_2} + D_{w_2} \widetilde{y} \end{cases} = \begin{cases} A_{w_2} x_{w_2} + B_{w_2} \mathrm{Cx} + B_{w_2} \mathrm{Du} \\ z = C_{w_2} x_{w_2} + D_{w_2} \mathrm{Cx} + D_{w_2} \mathrm{Du} \end{cases}$$

Din cate se poate vedea, $y = \tilde{y} + \tilde{w} = \mathrm{Cx} + \mathrm{Du} + C_{w_1} x_{w_1} + D_{w_1} w$. Daca scriem in state-space:

$$\begin{cases} \dot{x_{w_1}} = A_{w_1} x_{w_1} + B_{w_1} w \\ \tilde{w} = C_{w_1} x_{w_1} + D_{w_1} w \end{cases}$$

Acum strangem ecuatiile si rescriem:

$$\dot{x} = Ax + Bu$$
$$\dot{x}_{w_2} = B_{w_2} C x + A_{w_2} x_{w_2} + B_2 D u$$
$$\dot{x}_1 = A_{w_1} x_{w_1} + B_{w_1} w$$
$$z = D_{w_2} C x + C_{w_2} x_{w_2} + D_{w_2} D u$$
$$y = Cx + C_{w_1} x_{w_1} + D_{w_1} w + Du$$

In acest sens, putem scrie realizarea pe stare: $\begin{cases} \dot{x_T} = \begin{bmatrix} A & 0 & 0 \\ B_2 C & A_2 & 0 \\ 0 & 0 & A_1 \end{bmatrix} x_T + \begin{bmatrix} 0 & B \\ 0 & B_{w_2} D \\ B_{w_1} & 0 \end{bmatrix} u_T \\ y_T = \begin{bmatrix} D_{w_1} C & C_{w_2} & 0 \\ C & 0 & C_{w_1} \end{bmatrix} x_T + \begin{bmatrix} 0 & D_2 D \\ D_1 & 0 \end{bmatrix} u_T \end{cases}$, iar T va arata asa:

$$T = \left[\begin{array}{ccc|cc} A & 0 & 0 & 0 & B \\ B_{w_2} C & A_{w_2} & 0 & 0 & B_{w_2} D \\ 0 & 0 & A_{w_1} & B_{w_1} & 0 \\ \hline D_{w_2} C & C_{w_2} & 0 & 0 & D_2 D \\ C & 0 & C_{w_1} & D_1 & D \end{array}\right] = \left[\begin{array}{c|c} T_{11} & T_{12} \\ \hline T_{21} & T_{22} \end{array}\right]$$

Din cate stim, in TLFi, toate partile instabile vin in $T_{22} = \begin{bmatrix} A & 0 & 0 & B \\ B_{w_2}C & A_{w_2} & 0 & B_{w_2}D \\ 0 & 0 & A_{w_1} & 0 \\ C & 0 & C_{w_1} & D \end{bmatrix}$, unde $A_{22} = \begin{bmatrix} A & 0 & 0 \\ B_{w_2}C & A_{w_2} & 0 \\ 0 & 0 & A_{w_1} \end{bmatrix}$,

$B_{22} = \begin{bmatrix} B \\ B_{w_2}D \\ 0 \end{bmatrix}, C = \begin{bmatrix} C & 0 & C_{w_1} \end{bmatrix}, D = D.$

IMPORTANT: pana sa verificam daca sistemul este stabilizabil si detectabil, trebuie stiut ca scbimbul de coloane si/sau linii dintr-o matrice NU SCHIMBA RANGUL EI

Folosim PBH: $\begin{cases} \text{stabilizabil} \\ \text{detectabil} \end{cases} : \begin{cases} \text{rank} \begin{bmatrix} A - sI & 0 & 0 & B \\ B_{w_1}C & A_{w_2} - sI & 0 & B_2D \\ 0 & 0 & A_{w_1} - sI & 0 \end{bmatrix} \\ \text{rank} \begin{bmatrix} A - sI & 0 & 0 \\ B_{w_1}C & A_{w_2} - sI & 0 \\ 0 & 0 & A_{w_1} - sI \\ C & 0 & C_1 \end{bmatrix} \end{cases} = \begin{cases} \text{rank} \begin{bmatrix} A - sI & B & 0 & 0 \\ B_{w_1}C & B_2D & 0 & A_{w_2} - sI \\ 0 & 0 & A_{w_1} - sI & 0 \end{bmatrix} \\ \text{rank} \begin{bmatrix} A - sI & 0 & 0 \\ C & 0 & C_{w_1} \\ 0 & 0 & A_{w_1} - sI \\ B_{w_1}C & A_{w_2} - sI & 0 \end{bmatrix} \end{cases}.$

Vedem ca avem o matrice mai mica diagoala la fiecare matrice.

Pentru cazul de stabilizabilitate, rangul este dat pe linii. Matricea mica diagonala are rang $n_1 + n_2$. Noi stim ca G este minimal. Deci asta inseamna ca prima linie $[A - sI \quad B \quad 0 \quad 0]$ are rang intreg pe linii, adica n. In acest sens, rangul matricii este $n + n_1 + n_2$

Pentru cazul matricii de detectabilitate, rangul este dat pe coloane. Vedem matricea aia mica diagonala, deci rangul pe acele 2 coloane este $n_1 + n_2$. La fel, G minimal, deci rangul pe prima coloana $\begin{bmatrix} A - sI \\ C \\ 0 \\ 0 \end{bmatrix}$ are rang intreg pe linii, adica n. In acest sens, rangul matricii este $n + n_1 + n_2$

Sistem stabilizabil si detectabil, care are rangul $n + n_1 + n_2$

Mic flashback la TSA

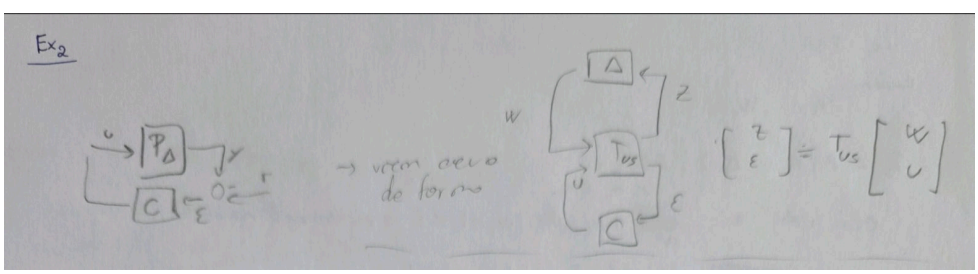
Noi aveam $\widetilde{P} = P(1 + W_T\Delta)$, unde $||\Delta||_\infty < 1$ si $\in \mathscr{RH}_\infty$. Noi trebuie sa facem un $C(s)$ stabil astfel incat $\sup_w(|W_S S| + |W_T T|) < 1$ pentru perfomanta robusta. Normal exista algoritmi pentur asta, dar nu sunt convecsi. De asemenea, trebuie sa ne asiguram de convergenta

Nu se poate ceva mai simplu? Ba da, si anume $\begin{cases} |W_S S| \\ |W_T T| \end{cases} < \frac{1}{2}$, relaxarea e excesiva. De ce? Noi practic spunem ca modelul sa fie de 2 ori mai bun fata de ce se cere, respectiv de 2 ori mai robust. Problema este ca sistemul nu este o solutie optima

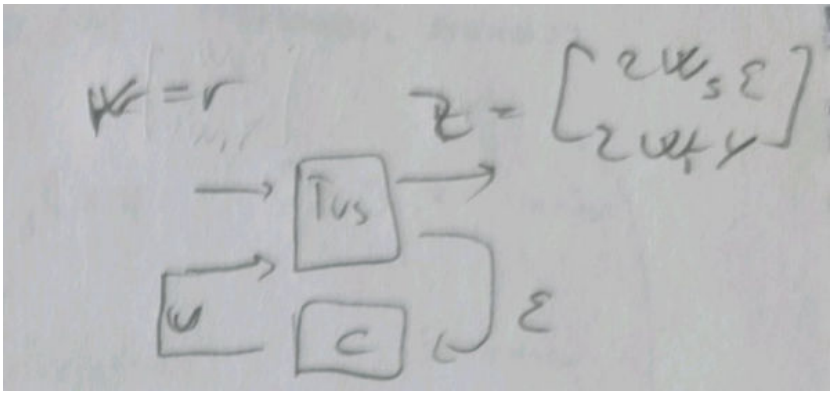
Cum scriem asta? $||\begin{bmatrix} 2W_S S \\ 2W_T T \end{bmatrix}||_\infty < 1 \Rightarrow ||\begin{bmatrix} W_S S \\ W_T T \end{bmatrix}||_\infty < \frac{1}{2} \Rightarrow \sqrt{|W_S S|^2 + |W_T T|^2} < \frac{1}{2} \Rightarrow |W_S S|^2 + |W_T T|^2 < \frac{1}{4}$, de unde scoatem $\begin{cases} |W_S S| < \frac{1}{2} \\ |W_T T| < \frac{1}{2} \end{cases}, \forall w$. Acum sa scriem problema de sensivitate mixta $||\begin{bmatrix} 2W_S S \\ 2W_T T \end{bmatrix}||_\infty < \gamma$. Uneori, daca este nevoie, se poate adauga si o limitare la comanda. Sistemul ar arata $||\begin{bmatrix} 2W_S S \\ 2W_T T \\ 2W_u u \end{bmatrix}||_\infty < \gamma$. Asta inseamna ca

$\begin{cases} ||S||_\infty \leq \gamma|(2W_S)^{-1}| \\ ||T||_\infty \leq \gamma|(2W_T)^{-1}| \\ ||\mathrm{KS}||_\infty \leq \gamma|(2W_u)^{-1}| \end{cases}$. Practic inseamna ca pe Bode noi vrem ca S, T, KS sa fie sub $2W_S, 2W_T$, respectiv $2W_u$

Haide sa vedem un exemplu:



Vrem $\min ||\begin{bmatrix} 2W_S S \\ 2W_T T \end{bmatrix}||_\infty$ si $\begin{bmatrix} 2W_S S \\ 2W_T T \end{bmatrix} = \mathrm{TFLi}(T, C)$



Acum vine o intrebare naturala: **Cine naiba sunt z si w?**

Hai sa aflam

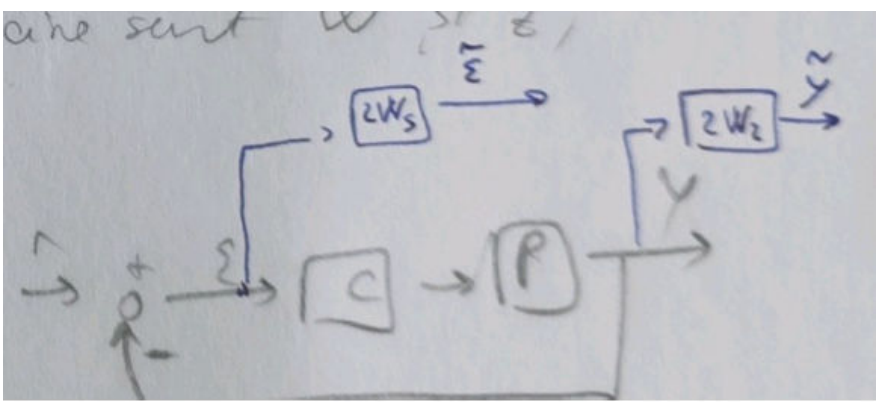
$\begin{cases} \epsilon = \mathrm{Sr} \\ y = \mathrm{Tr} \end{cases}$, unde S este functia de sensibilitate: $T_{\mathrm{er}} = \dfrac{1}{1+L}$, iar T este sistemul in bucla inchisa: $T_{\mathrm{yr}} = \dfrac{L}{1+L}$.

Inmultim prima egalitate cu $2W_S$, iar pe a doua cu $2W_T$. Obtinem $\begin{cases} 2W_S\epsilon = 2W_S\mathrm{Sr} \\ 2W_T y = 2W_T\mathrm{Tr} \end{cases}$. Sa scriem in state-space:

$z = \begin{bmatrix} 2W_S\epsilon \\ 2W_T y \end{bmatrix} = \begin{bmatrix} 2W_S S \\ 2W_T T \end{bmatrix} r$, $r = w$. Avem $\begin{bmatrix} 2W_S S \\ 2W_T T \end{bmatrix}$ si vrem sa scriem ca TLFi.

Pas1: gasirea lui w si z

$$\begin{bmatrix} 2W_S\epsilon \\ 2W_T y \\ \epsilon \end{bmatrix} = T_{\mu s} \begin{bmatrix} r \\ u \end{bmatrix}$$



De aici scoatem: $\begin{cases} y = \mathrm{Pu} \\ \tilde{y} = 2W_T y = 2W_T \mathrm{Pu} \\ \epsilon = r - y = r - \mathrm{Pu} \\ \tilde{\epsilon} = 2W_S \epsilon = 2W_S(r - \mathrm{Pu}) \end{cases}$

Acum calculam realizari minimale pentru $\begin{cases} G = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \\ W_S = \begin{bmatrix} A_S & B_S \\ C_S & D_S \end{bmatrix} \\ W_T = \begin{bmatrix} A_T & B_T \\ C_T & D_T \end{bmatrix} \end{cases}$

$$\begin{cases} \dot{x}(t) = \mathrm{Ax}(t) + \mathrm{Bu}(t) \\ y(t) = \mathrm{Cx}(t) + \mathrm{Du}(t) \end{cases}$$

$$\begin{cases} \dot{x_T}(t) = A_T x_T(t) + B_T y(t) = A_T x_T + B_T \mathrm{Cx}(t) + B_T \mathrm{Du}(t) \\ \tilde{y}(t) = 2C_T x_T(t) + 2D_T y(t) = C_T x_T(t) + 2D_T \mathrm{Cx}(t) + 2D_T \mathrm{Du}(t) \end{cases}$$

$\begin{cases} \dot{x_S}(t) = A_S x_S(t) + B_S \epsilon = A_S x_S(t) - B_S r(t) - B_S \mathrm{Cx}(t) - B_S \mathrm{Du}(t) \\ \tilde{\epsilon}(t) = 2C_S x_S(t) + 2D_S \epsilon = 2C_S x_S(t) + 2D_S \mathrm{Cx}(t) + 2D_S r(t) - 2D_S \mathrm{Du}(t) \end{cases}$. Acum sa rescriem mai frumos:

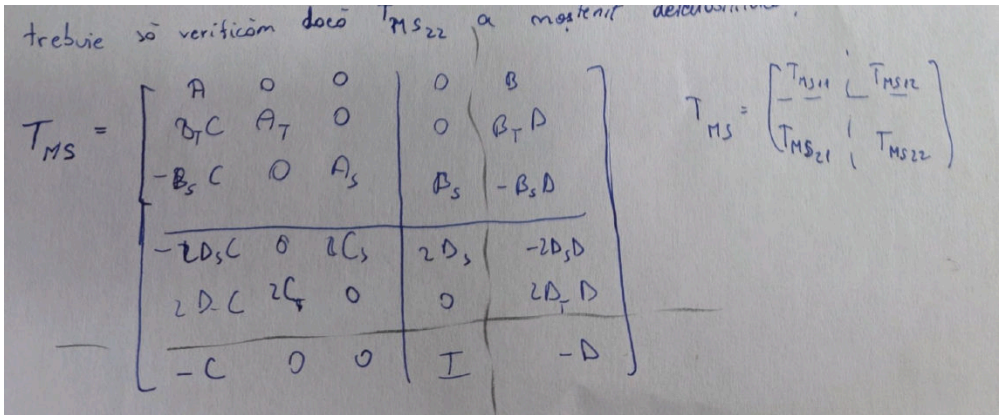
$$\dot{x} = Ax + Bu$$
$$\dot{x}_T = B_T C x + A_T x_T + B_T D u$$
$$\dot{x}_s = -B_s C x + A_s x_s + B_s r - B_s D u$$
$$\tilde{\varepsilon} = -2D_s C x + 2C_s x_s + 2D_s r - 2D_s D u$$
$$\tilde{y} = 2D_T C x + 2C_T x_T + 2D_T D u$$
$$\varepsilon = -Cx + r - Du$$

Scoatem: $x_{\mathrm{ms}} = \begin{bmatrix} x \\ x_T \\ x_S \end{bmatrix}, u_{\mathrm{ms}} = \begin{bmatrix} r \\ u \end{bmatrix}, y_{\mathrm{ms}} = \begin{bmatrix} \tilde{\epsilon} \\ \tilde{y} \\ \epsilon \end{bmatrix}$

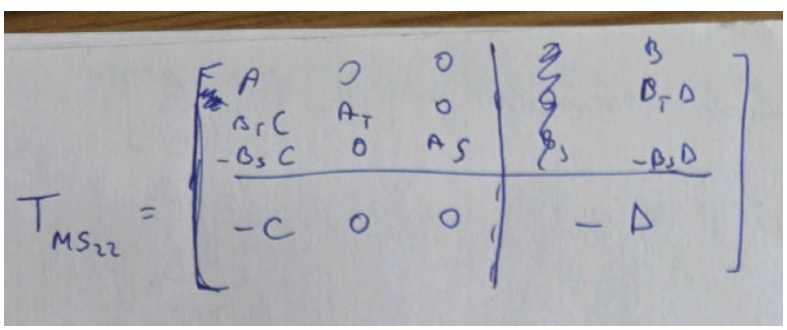
Acum sa scriem in state-space:

$$\begin{cases} \dot{x}_{\text{ms}} = \begin{bmatrix} A & 0 & 0 \\ B_T C & A_T & 0 \\ -B_S C & 0 & A_S \end{bmatrix} x_{\text{ms}} + \begin{bmatrix} 0 & B \\ 0 & B_T D \\ B_S & -B_S D \end{bmatrix} u_{\text{ms}} \\ y_{\text{ms}} = \begin{bmatrix} -2D_S C & 0 & 2C_S \\ 2D_T C & 2C_T & 0 \\ -C & 0 & 0 \end{bmatrix} x_{\text{ms}} + \begin{bmatrix} 2D_S & -2D_S D \\ 0 & 2D_T D \\ I & -D \end{bmatrix} u_{\text{ms}} \end{cases}$$

. Acum compunem pe T_{ms} si vedem daca $T_{\text{ms}_{22}}$ mosteneste detectabilitatea si stabilitatea



$$T_{\text{ms}_{22}} = \begin{bmatrix} A & 0 & 0 & B \\ B_T C & A_T & 0 & B_T D \\ -B_S C & 0 & A_S & -B_S D \\ -C & 0 & 0 & -D \end{bmatrix}$$



Ne uitam la $A_{\text{ms}_{22}}$. Vedem ca are forma diagonala, deci valorile proprii sunt reuniunea celor 3 matrici de pe diagonala principala. **Reamintim ca P, W_S, W_T le-am luat minimale**

Pentru ca solutia sa mearga, trebuie ca $\begin{cases} T_{\mathrm{ms}_{12}}(\infty) \text{ sa aiba rang intreg pe coloane} \Leftrightarrow D \neq 0 \\ T_{\mathrm{ms}_{21}}(\infty) \text{ sa aiba rang intreg pe linii. Asta se respecta} \end{cases}$. De ce? Sa ne uitam la $T_{\mathrm{ms}_{12}}$. La $D_{\mathrm{ms}_{12}} = \begin{bmatrix} -2D_S D \\ -2D_T D \end{bmatrix}$. Daca D = 0, nasol. Pica rangul

```
clear
close all
clc

s = tf('s');

G = 1 / (s + 1);

M_s = 4.5;
w_B = 2;
A_s = 1e-4;
Ws = (s/M_s + w_B) / (s + w_B*A_s);

M_t = 3.5;
w_bt = 20;
A_t = 1e-3;
Wt = (s + w_bt/M_t) / (A_t*s + w_bt);


Ws_mod = 2 * Ws;
Wt_mod = 2 * Wt;

help mixsyn
```

```
--- help for DynamicSystem/mixsyn ---

 DynamicSystem/mixsyn - Mixed-sensitivity H_{∞} synthesis method for robust control loop-shaping design
    This MATLAB function computes a controller that minimizes the H∞ norm of
    the weighted closed-loop transfer function

    Syntax
      [K,CL,gamma,info] = mixsyn(G,W1,W2,W3)
      [K,CL,gamma] = mixsyn(G,W1,W2,W3,gamTry)
      [K,CL,gamma] = mixsyn(G,W1,W2,W3,gamRange)
      [K,CL,gamma] = mixsyn(___,opts)
      [K,CL,gamma,info] = mixsyn(___)

    Input Arguments
      G - Plant
        dynamic system model
      W1,W2,W3 - Weighting functions
        dynamic system model | []
      gamTry - Target performance level
        positive scalar
      gamRange - Performance range for search
        [0,Inf] (default) | vector of form [gmin,gmax]
      opts - Options
```

```
        hinfsynOptions object

    Output Arguments
      K - Controller
        ss model object | []
      CL - Augmented closed-loop transfer function
        ss model object | []
      gamma - Controller performance
        nonnegative scalar | Inf
      info - Synthesis data
        structure | []

    Examples
      Loop Shaping with mixsyn

    See also augw, hinfsyn, makeweight, hinfsynOptions

    Introduced in Robust Control Toolbox before R2006a
    Documentation for DynamicSystem/mixsyn
```

```
[K, CL, gamma, info] = mixsyn(G, Ws_mod, [], Wt_mod);
```

Golul ala acolo este pentru limitare la comanda: $2W_u u$

G este functia/matricea de transfer, Ws_mod este limitare la eroare, iar Wt_mod este limitare la iesire

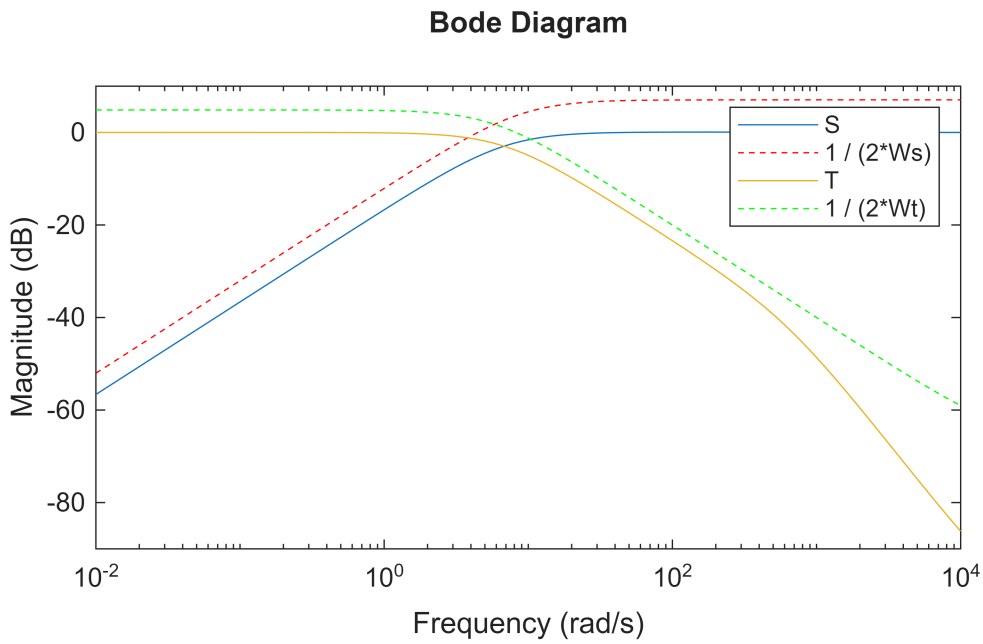
```
fprintf('Valoare gamma: %.4f\n', gamma);
```

```
Valoare gamma: 0.8210
```

```
if gamma < 1
    disp('Merge. Criteriul || [2WsS; 2WtT] ||_infinit < 1 e ok!');
else
    disp('Nu e ok');
end
```

```
Merge. Criteriul || [2WsS; 2WtT] ||_infinit < 1 e ok!
```

```
S = feedback(1, G*K);
T = feedback(G*K, 1);

figure;
bodemag(S, 1/Ws_mod, 'r--', T, 1/Wt_mod, 'g--', {1e-2, 1e4});
legend('S', '1 / (2*Ws)', 'T', '1 / (2*Wt)');
```



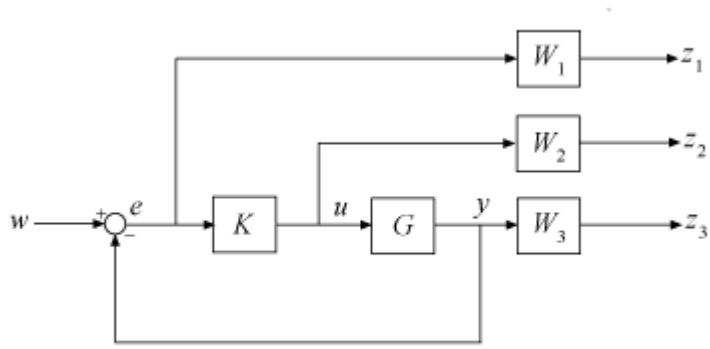
Din ce inteleg, ne uitam la Bode pentru a ne asigura ca S este sub $\frac{1}{|2W_S|}$, respectiv T sa fie sub $\frac{1}{|2W_T|}$

De aici am citit:

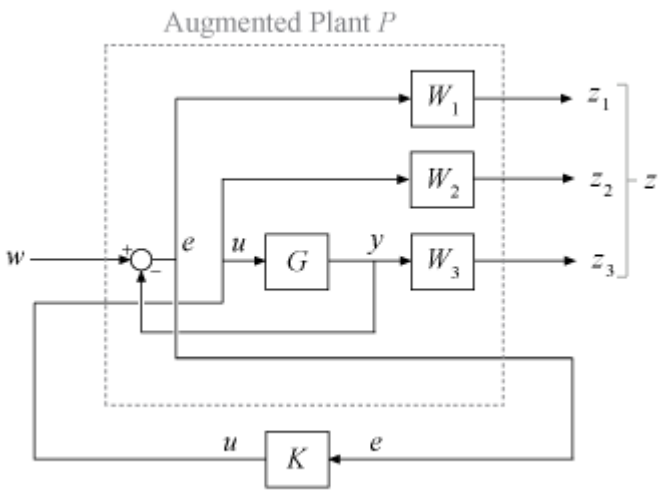
https://www.mathworks.com/help/releases/R2025b/robust/gs/using-mixsyn-for-h-infinity-loop-shaping.html

Rezumat la link:

Aici punem $2W_S$ la eroare, $2W_T$ la iesire si $2W_u$ la intrare. Matlab fac fara acel 2. Bagati voi 2 "mental"



Dupa aceea, reordoneaza astea pentru a obtine un TLFi (T,K). Ei noteaza P



Daca scriem matricea de transfer de la w la z: $\begin{bmatrix} W_S S \\ W_T T \\ W_u \mathrm{KS} \end{bmatrix}$, unde $\begin{cases} S = (I + \mathrm{GK})^{-1} \\ T = \mathrm{GK}(I + \mathrm{GK})^{-1} \\ \mathrm{KS} = K(I + \mathrm{GK})^{-1} \end{cases}$